

W01 WATER QUALITY INDICATORS | P

Intent : Verify the quality of water for human contact through easy-to-test parameters.

Summary : This WELL feature requires the provision of water that meets thresholds for turbidity and coliforms for all water likely to come in contact with building occupants and verifies performance using on-site tests.

Issue : Most cities have an extensive treatment system to produce and deliver water with safety and integrity. Two parameters—total coliforms and turbidity— are commonly used to assess the effectiveness of these systems as easy-to-test measurements for the possible presence of other, more concerning contaminants, and are therefore known as 'indicators'. Coliform bacteria are naturally present in the environment and are generally considered harmless. However, some coliforms are associated with fecal contamination and may cause disease if ingested.¹ Turbidity is a measure of water cloudiness, which per-se does not constitute a health concern, however, it does relate to the availability of food and shelter for microbes, the presence of particulate contaminants and issues with the water treatment process,² along with posing an aesthetic concern. High turbidity water also can reduce the efficacy of water treatment technologies.¹

Solutions : Water filtration can reduce turbidity and, depending on the type of device, may also trap bacteria and other contaminants. If the water has low turbidity, disinfection at the point of use with ultraviolet (UV) light may be effective at killing coliforms and pathogenic microbes.²

PART 1 VERIFY WATER QUALITY INDICATORS

For All Spaces:

Water delivered to the project and intended for human contact (e.g., drinking, cooking and dishwashing, handwashing, showering or bathing) meets the following thresholds:

- a. Turbidity is less than or equal to 1.0 NTU, FTU or FNU (nephelometric turbidity, formazin turbidity or formazin nephelometric units, respectively).
- b. Coliforms are not detected in any 100 ml sample.

Note : Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. US Centers for Disease Control & Prevention. A Guide to Drinking Water Treatment Technologies for Household Use. [Reference](#). Published 2014. Accessed December 10, 2019.

W02 DRINKING WATER QUALITY | P

Intent : Provide access to drinking water that complies with health-based limits on chemical composition.

Summary : This WELL feature requires projects to provide drinking water that meets thresholds on chemicals as published by research and regulatory organizations.

Issue : The chemical composition of drinking water, and therefore, its quality, changes from city to city and even within buildings due to the highly variable conditions of its sourcing, treatment and distribution within cities and inside buildings.¹ For example, natural deposits have caused arsenic to leach into some groundwaters to reach levels above drinking water health guidelines.² Water streams can also pick up contaminants from agricultural runoffs and direct industrial discharges,³ whereas drinking water may encounter many opportunities to pick up contaminants in its travel from a treatment plant to the point of use, including corrosion byproducts such as lead and copper.⁴ Disinfectants used to prevent microbial growth and render water potable, such as chlorine, may react with natural organic matter and yield unwanted disinfectant byproducts (DBPs) such as trihalomethanes (THMs) and haloacetic acids (HAAs), to which chronic exposure needs to be minimized.⁴

Solutions : Drinking water is treated and distributed to meet applicable legal requirements and regulations that may differ by country,⁵ and many building-scale interventions can improve water quality depending on the contaminants that need to be removed. Typical technologies able to capture contaminants include activated carbon filters, ion exchange resins and reverse osmosis (RO) systems. Evaluating chemical parameters such as pH and free chlorine may inform of the potential for the uptake of corrosion byproducts and/or bacterial growth in drinking water.¹

PART 1 MEET CHEMICAL THRESHOLDS

For All Spaces:

The following requirements are met:

a.

The project provides at least one drinking water dispenser, plus one drinking water dispenser per dwelling unit (if applicable).

b. Drinking water dispensers provide water that meets the following parameters:¹

1. Arsenic ≤ 0.01 mg/L.
2. Cadmium ≤ 0.003 mg/L.
3. Chromium (total) ≤ 0.05 mg/L.
4. Copper ≤ 2 mg/L.
5. Fluoride ≤ 1.5 mg/L.
6. Lead ≤ 0.01 mg/L.
7. Mercury (total) ≤ 0.006 mg/L.
8. Nickel ≤ 0.07 mg/L.
9. Nitrate ≤ 50 mg/L as Nitrate (11 mg/L as Nitrogen).
10. Nitrite ≤ 3 mg/L as Nitrite (0.9 mg/L as Nitrogen).
11. Total chlorine ≤ 5 mg/L.

c. Drinking water dispensers provide water that meets the following parameters:

1. Residual (free) chlorine does not exceed 4 mg/L.³
2. The concentration of total trihalomethanes (TTHM, sum of dibromochloromethane, bromodichloromethane, chloroform and bromoform) is 0.08 mg/L or less.³
3. The concentration of haloacetic acids (HAA5, sum of chloroacetic, dichloroacetic, trichloroacetic, bromoacetic and dibromoacetic acids) is 0.06 mg/L or less.³

Note : Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET THRESHOLDS FOR ORGANICS AND PESTICIDES

For All Spaces:

Option 1: Drinking water quality report

The following requirements are met:

a. A municipal water quality report issued not more than one year before enrollment or the start of subscription reports on at least two of the pesticides below. All reported pesticides within the municipal water quality report comply with the following thresholds or are found to be compliant through Option 2, On-Site Testing:¹

1. Aldrin and Dieldrin (combined): 0.00003 mg/L or less.
2. Atrazine: 0.1 mg/L or less.
3. Carbofuran: 0.007 mg/L or less.
4. Chlordane: 0.0002 mg/L or less.
5. 2,4-Dichlorophenoxyacetic acid (2,4-D): 0.03 mg/L or less.
6. Dichlorodiphenyltrichloroethane (DDT) and metabolites: 0.001 mg/L or less.

7. Lindane: 0.002 mg/L or less.
8. Pentachlorophenol (PCP): 0.009 mg/L or less.
- b. A municipal water quality report issued not more than one year before enrollment or the start of subscription reports on at least three of the organic contaminants below. All reported organic contaminants within the municipal water quality report comply with the following thresholds or are found compliant through Option 2, On-Site Testing:¹
 1. Benzene: 0.01 mg/L.
 2. Benzo[a]pyrene: 0.0007 mg/L.
 3. Carbon tetrachloride: 0.004 mg/L.
 4. 1,2-Dichloroethane: 0.03 mg/L.
 5. Tetrachloroethene (Tetrachloroethylene): 0.04 mg/L.
 6. Toluene: 0.7 mg/L.
 7. Trichloroethene: 0.02 mg/L.
 8. 2,4,6-Trichlorophenol: 0.2 mg/L.
 9. Vinyl Chloride: 0.0003 mg/L.
 10. Xylenes (o-, m- and p-): 0.5 mg/L.

OR

Option 2: On-site testing

The following requirements are met:

- a. Testing must be conducted at drinking water dispensers for at least two of the pesticides from the list below. All contaminants in the list below that are reported by the lab must comply with the following thresholds:
 1. Aldrin and Dieldrin (combined): 0.00003 mg/L or less.
 2. Atrazine: 0.1 mg/L or less.
 3. Carbofuran: 0.007 mg/L or less.
 4. Chlordane: 0.0002 mg/L or less.
 5. 2,4-Dichlorophenoxyacetic acid (2,4-D): 0.03 mg/L or less.
 6. Dichlorodiphenyltrichloroethane (DDT) and metabolites: 0.001 mg/L or less.
 7. Lindane: 0.002 mg/L or less.
 8. Pentachlorophenol (PCP): 0.009 mg/L or less.
- b. Testing must be conducted at drinking water dispensers for at least three of the organic contaminants from the list below. All contaminants in the list below that are reported by the lab must comply with the following thresholds:
 1. Benzene: 0.01 mg/L.
 2. Benzo[a]pyrene: 0.0007 mg/L.
 3. Carbon tetrachloride: 0.004 mg/L.
 4. 1,2-Dichloroethane: 0.03 mg/L.
 5. Tetrachloroethene (Tetrachloroethylene): 0.04 mg/L.
 6. Toluene: 0.7 mg/L.
 7. Trichloroethene (Trichloroethylene): 0.02 mg/L.
 8. 2,4,6-Trichlorophenol: 0.2 mg/L.
 9. Vinyl Chloride: 0.0003 mg/L.
 10. Xylenes (o-, m- and p-): 0.5 mg/L.

Note: Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. World Health Organization. Arsenic. <http://www.who.int/news-room/fact-sheets/detail/arsenic>. Published 2018. Accessed 2020.
3. U.S. Environmental Protection Agency. National primary drinking water regulations. <https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-regulations>. Published 2009. Accessed January 20, 2020.

4. World Health Organization. Trihalomethanes in Drinking-water. Geneva, Switzerland 2005.
5. World Health Organization. Water Safety in Buildings. Geneva, 2011.

W03 BASIC WATER MANAGEMENT | P

Intent : Implement protocols to reduce risk of water quality loss and *Legionella* colonization

Summary : This WELL feature requires projects to proactively test drinking water and to manage recirculating hot water systems against *Legionella* colonization.

Issue : All water systems require some degree of validation to ensure that the health targets are met under its operational conditions, for which sampling is an effective verification tool ¹. *Legionella* bacteria is naturally present in waters at low concentrations, but it may colonize recirculated water systems and can cause lung disease and even death if contaminated water aerosols are inhaled.² Legionnaire's disease affects especially the immunocompromised, smokers and those over age 50. ² Assets commonly vulnerable to *Legionella* infestation include domestic potable and hot water systems, cooling towers, humidifiers, misters, decorative fountains, spas and hot tubs.²

Solutions : Basic management for water quality ensures that the water sourcing, treatment and delivery operates as designed. Turbidity, residual chlorine and pH monitoring help control basic chemical and microbiological water characteristics and, if trends are detected, may inform that actions may be needed to protect the quality of the water, such as changing a filter, check the building's pipes for leaks or inquire with the city for major works in the area. Management for minimizing *Legionella* requires a thorough and quantitative risk assessment of buildings' water assets, identifying locations where control is required, and a well-documented maintenance and operations program.³ Implementing a proper *Legionella* management plan should reduce the risk of exposure to pathogenic bacteria.

PART 1 MONITOR CHEMICAL AND BIOLOGICAL WATER QUALITY

For All Spaces except Dwelling Units:

1: Drinking water quality report

The following requirements are met:

- a. The following water parameters are sampled at drinking water dispensers in occupiable spaces at intervals of no less than once per year:
 1. Turbidity.
 2. pH.
 3. Residual (free) chlorine.
 4. Total coliforms, only if residual chlorine is below detection limits
- b. Tests are required at 5% of drinking water dispensers, up to a maximum of four tests.
- c. The water quality results are submitted annually through the WELL digital platform.

Note :

On-going testing does not need to be performed by a WELL Performance Testing Agent.

PART 2 IMPLEMENT LEGIONELLA MANAGEMENT PLAN

For All Spaces:

1: Legionella plan development

The project provides a Legionella management plan that meets the following requirements:

- a. Addresses hot water systems, cooling towers, decorative fountains and any other devices or spaces under control of the project where water is recirculated and aerosolized.
- b. Includes the items listed below:³
 1. Determination of roles for Legionella management in the building, distinguishing those under project control from those that may be the responsibility of building management or other parties.
 2. Water system inventory and process flow diagrams of systems within the project boundary.
 3. Hazard analysis of water assets within the project boundary. If the project does not operate the building hot water supply system (e.g., boilers, heaters, pumps or hot water risers), then an explanation of the building-wide Legionella management policies (if any) and how they influence risk is included.
 4. A list of monitoring actions for relevant variables (e.g., temperature or residual chlorine), performance limits associated with these variables, and corrective actions when variables exceed such limits.
 5. A list of critical control points (locations where actions to maintain relevant variables listed in (4) within performance limits are applied) within the project boundary.
 6. Verification and validation procedures for evaluating the suitability and proper implementation of the management plan. A Legionella sampling schedule is included if projects have operational control over cooling towers and spas.
 7. Protocols for documenting results of monitoring activities and corrective actions. If sampling for Legionella is planned, results are included.

2: Legionella plan implementation

The following requirement is met:

- a. Project submits annually through the WELL digital platform documentation of monitoring results, corrective actions and *Legionella* sample results (if any) as stated in the Legionella management plan.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. US Centers for Disease Control & Prevention. Legionella (Legionnaires' Disease and Pontiac Fever). <https://www.cdc.gov/legionella/index.html>. Published 2019. Updated April 30, 2018. Accessed December 20, 2019.
3. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 188-2018: Legionellosis: Risk Management for Building Water Systems. In.

W04 ENHANCED WATER QUALITY | O (MAX : 1 PT)

Intent : Provide access to drinking water without unpleasant taste, odor and appearance.

Summary : This WELL feature requires projects to provide drinking water that meets thresholds on chemicals that affect aesthetics and taste concerns.

Issue : Even when health-based thresholds for water quality are met, water may be found unappealing to drink because of taste, odor and appearance concerns. For example, high levels of chloride contribute to a salty taste and iron can give the water a reddish appearance.¹ Therefore, some regulatory bodies set non-enforceable limits based on human detectability and acceptability for these substances.^{2,3}

Solutions : Like pollutants with health-based concerns, the treatment system to address nuisance chemicals depends on the contaminant of interest. Treatment options include filtration with carbon media and reverse osmosis.

Impact : By managing nuisance chemicals, projects can provide more appealing and palatable drinking water.

PART 1 MEET THRESHOLDS FOR DRINKING WATER TASTE (MAX : 1 PT)

For All Spaces:

Water delivered to the project for human consumption meets the following thresholds:

- a. Aluminum ≤ 0.2 mg/L.²
- b. Chloride ≤ 250 mg/L.²
- c. Copper ≤ 1 mg/L.²
- d. Manganese ≤ 0.05 mg/L.
- e. Iron ≤ 0.3 mg/L.²
- f. Silver ≤ 0.1 mg/L.²
- g. Sodium ≤ 270 mg/L.³
- h. Sulfate ≤ 250 mg/L.²
- i. Zinc ≤ 5 mg/L.²
- j. Total Dissolved Solids (TDS) ≤ 500 mg/L.²
- k. Free Chlorine ≤ 1.25 mg/L.⁴

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.
2. U.S. Environmental Protection Agency. National Secondary Drinking Water Regulations. <https://www.epa.gov/dwstandardsregulations/secondary-drinking-water-standards-guidance- nuisance-chemicals>. Published 2017. Accessed.
3. Health Canada. Guidelines for Canadian Drinking Water Quality—Summary Table. <https://www.canada.ca/en/health-canada/services/environmental-workplace-health/reports-publications/water-quality/guidelines-canadian-drinking-water-quality-summary-table.html>. Published 2017. Accessed.
4. Crider Y, Sultana S, Unicomb L, Davis J, Luby SP, Pickering AJ. Can you taste it? Taste detection and acceptability thresholds for chlorine residual in drinking water in Dhaka, Bangladesh. Science of The Total Environment. 2018;613-614:840-846.

W05 DRINKING WATER QUALITY MANAGEMENT | O (MAX : 3 PT)

Intent : Maintain and display consistent high quality of drinking water.

Summary : This WELL feature requires pre-testing of water quality parameters to determine treatment needs, monitoring at a more frequent interval and disclosure of water results.

Issue : Providing potable water to buildings is a multi-stage process that involves sourcing of the water, potabilization in treatment plants, distribution through network of pipes and delivery to the tap. While steady delivery of potable water is a reality in many places, other places must regularly contend with water that is delivered below potability standards or with fluctuating quality due to the intrusion of contaminants in the water distribution pipes,¹ unsupervised changes in municipal water supply and treatment² or weather-related events.

Solutions : From a building perspective, sound water quality management begins with an understanding of the incoming water quality, preferably through testing and analysis of historical data. If needed, treatment devices, such as filters or UV disinfection units, can be used to achieve data-driven, health-based water quality targets.³ Periodic water monitoring not only confirms the quality of the water, but also helps to determine the needs for maintenance in pipes, fixtures or treatment devices. Availability of water quality results and maintenance records to occupants may also increase drinking water consumption, furthering both cost-saving and sustainability efforts for the project, while also promoting occupant hydration.

PART 1 ASSESS AND MAINTAIN DRINKING WATER QUALITY (MAX : 2 PT)

For All Spaces:

1: Water quality pre-test

For first-time registered projects, the following requirements are met:

- a. The project pre-tests water at least one month before Performance Verification for the parameters below:
 1. Turbidity.
 2. Coliforms.
 3. pH.
 4. Total Dissolved Solids (TDS).
 5. Total Chlorine.
 6. Residual (free) chlorine.
 7. Arsenic.
 8. Lead.
 9. Copper.
 10. Nitrate
 11. Benzene.
- b. Sampling occurs at the following locations:
 1. The water dispenser that is closest to the pipe that delivers water into the project, before any point-of-entry water treatment system where possible.
 2. For projects with more than two floors or more than 930 m² in area, a second drinking water dispenser on the highest floor to which the project has access that is farthest from the location in requirement b(1).
 3. For projects of 12 or more floors, one additional drinking water dispenser for every 10 floors.
- c. Samples must be taken with point-of-use filters or other water treatment devices bypassed or removed, if present.

Note : Projects pursuing re-certification do not need to achieve Option 1.

2: Water quality monitoring

The following requirements are met:

- a. Piped water is delivered to drinking water dispensers.
- b. Water is tested quarterly in drinking water dispensers and meets the following thresholds. If any sample exceeds these thresholds, remediation and re-testing occur within a month:
 1. Turbidity is 1.0 NTU, FTU or FNU or less.
 2. pH is between 6.5 and 9.0 (between 5.5 and 9 if a reverse osmosis system is installed at the point of use).
 3. Total Dissolved Solids (TDS) are 500 mg/L or less.
 4. Total Chlorine is 5 mg/L or less.
 5. Residual (free) Chlorine is 5 mg/L or less.
 6. Total Coliforms are not detected in a 100 ml sample. Testing is required only if residual chlorine is not detected.
 7. Lead is 10 µg/L or less. Sampling frequency can be reduced to once per year if results are below detection limits in two consecutive samples.
 8. Copper is 2 mg/L or less. Sampling frequency can be reduced to twice a year if results are below 0.1 mg/L in two consecutive samples; testing is no longer required if four consecutive samples are below this threshold.
- c. The number and location of sampling points for on-going monitoring complies with the requirements outlined in

the Performance Verification Guidebook. For pH, use sampling locations and frequency set for residual chlorine.

d. All test results are submitted annually through the WELL digital platform.

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 PROMOTE DRINKING WATER TRANSPARENCY (MAX : 1 PT)

For All Spaces:

The following information is prominently displayed near sources of drinking water (or on a website available to occupants):

- a. Water quality results from the most recent sampling, including date of testing and compliance with WELL thresholds.
- b. If filters or other treatment units are in use, information about the treatment technologies and most recent date of device maintenance and/or filter cartridge replacement.

REFERENCES

1. World Health Organization. Water Safety in Buildings. Geneva, 2011.
2. Torrice M. How Lead Ended Up In Flint's Water. C&EN Global Enterprise. 2016;94(7):26-29.
3. World Health Organization. Guidelines for drinking-water quality. 4th ed. Geneva, Switzerland: WHO Press; 2017.

W06 DRINKING WATER PROMOTION | O (MAX : 1 PT)

Intent : Promote proper hydration through the consumption of drinking water over less healthy alternatives by promoting access to drinking water of verified quality.

Summary : This WELL feature requires readily available and maintained dispensers for drinking water.

Issue : People in many parts of the world do not hydrate enough to meet health guidelines, even in places where drinking water is available at the tap such as the U.S., Europe and Australia.¹⁻³ Conversely, increased amounts of water consumption has been connected with better diet quality.²⁻⁴ Despite being relatively cheap and being shown to attenuate socioeconomic disparities in hydration,⁵ the public's negative perceptions about the quality of tap water has catalyzed the adoption of bottled water. While plastic bottled water allows for portable hydration, excess waste produced by the dumping and degradation of single-use bottles and the generation of microplastics is a serious environmental concern. Pollution produced by single use plastic products is a rising public health issue that requires a comprehensive toxicological assessment in order to understand its full impact.⁶

Solutions : The first step toward increasing consumption of good quality water is to make it easily available and removing barriers to accessibility. When the addition of drinking fountains is combined with information sessions, consumption of plain water has been found to increase.⁷ Finally, it is important to keep water dispensers in a state of good repair to encourage continued water consumption.⁸ Using drinking water fixtures also allows fresh water to run through pipes, preventing losses in water quality.⁹

PART 1 ENSURE DRINKING WATER ACCESS (MAX : 1 PT)

For All Spaces except Dwelling Units:

1: Dispenser availability

The following requirements are met:

- a. At least one drinking water dispenser on each floor that includes regularly occupied space is located within a 30 m walk distance of all regularly occupied floor area and in all dining areas (except in areas where the presence of drinking water dispensers is forbidden by building codes or applicable regulations).
- b. Water delivered by the dispensers is directly piped through the building's water supply or is stored in containers designed for refilling.
- c. All newly installed drinking water fountains are designed for water bottle-refilling.

2: Dispenser maintenance

All drinking water dispensers meet the following requirement:

- a. The mouthpieces/outlets, protective guards, aerators (if present), basins and touch points are cleaned on a daily basis.

REFERENCES

1. Malisova O, Athanasatou A, Pepa A, et al. Water Intake and Hydration Indices in Healthy European Adults: The European Hydration Research Study (EHRS). *Nutrients*. 2016;8(4).
2. Sui Z, Zheng M, Zhang M, Rangan A. Water and Beverage Consumption: Analysis of the Australian 2011-2012 National Nutrition and Physical Activity Survey. *Nutrients*. 2016;8(11).
3. Kenney EL, Long MW, Craddock AL, Gortmaker SL. Prevalence of Inadequate Hydration Among US Children and Disparities by Gender and Race/Ethnicity: National Health and Nutrition Examination Survey, 2009–2012. *American Journal of Public Health*. 2015;105(8):e113-e118.
4. Gazan R, Sondey J, Maillot M, Guelinckx I, Lluich A. Drinking Water Intake Is Associated with Higher Diet Quality among French Adults. *Nutrients*. 2016;8(11):689.
5. Brooks CJ, Gortmaker SL, Long MW, Craddock AL, Kenney EL. Racial/Ethnic and Socioeconomic Disparities in Hydration Status Among US Adults and the Role of Tap Water and Other Beverage Intake. *American Journal of Public Health*. 2017;107(9):1387-1394.
6. Picó Y, Barceló D. Analysis and Prevention of Microplastics Pollution in Water: Current Perspectives and Future Directions. *ACS Omega*. 2019;4(4):6709-6719.
7. Jasper C, Le T-T, Bartram J. Water and Sanitation in Schools: A Systematic Review of the Health and Educational Outcomes. *International Journal of Environmental Research and Public Health*. 2012;9(8):2772-2787.
8. Patel AI, Bogart LM, Uyeda KE, Rabin A, Schuster MA. Perceptions about availability and adequacy of drinking water in a large California school district. *Prev Chronic Dis*. 2010;7(2):A39-A39.
9. Ling F, Whitaker R, LeChevallier MW, Liu W-T. Drinking water microbiome assembly induced by water stagnation. *The ISME Journal*. 2018;12(6):1520-1531.

W07 MOISTURE MANAGEMENT | O (MAX : 3 PT)

Intent : Limit the potential for bacteria and mold growth within buildings from water infiltration, condensation and internal leaks.

Summary : This WELL feature requires projects to develop strategies to minimize the presence of unintentional water and, when unavoidable, to manage it through material selection and inspections.

Issue : Excess moisture and dampness is a common problem in buildings, affecting about 20% of buildings in Europe, Canada and the U.S.¹ When improperly managed, moisture creates conditions conducive to the growth of mold and other biological pests, which can increase the risk of developing respiratory infections and asthma for those within the building.² It is estimated that one fifth of asthma cases in the U.S. are caused by excess moisture and dampness in buildings.² Furthermore, moisture can damage the building itself by creating an environment hospitable to insects and other destructive pests, corroding metal components and degrading wood and porous building materials.³ These problems can arise when water unintentionally penetrates the building envelope or leaks from indoor uses of water, or when moisture-heavy air condenses on building materials.

Solutions : Through effective design of the building's curtain wall, water piping assemblies and ventilation systems, and by selecting appropriate materials where condensation may occur, projects can make conditions inhospitable to mold, microbes and pests, reducing the risk to respiratory health.^{3,4} Inspections are needed to both verify that design and operations properly safeguards against mold growth, as well as inform the need for preventative maintenance.⁴

PART 1 DESIGN ENVELOPE FOR MOISTURE PROTECTION (MAX : 1 PT)

For All Spaces:

The building envelope aims to minimize moisture intrusion and accumulation through the following:³

- a. For a project where construction occurs after enrollment or the start of subscription, verification of site drainage and storm water management during building construction phase.
- b. Air tightness testing to assess water vapor transfer.
- c. Adverse vapor pressure differentials that may cause condensation on interstitially hidden materials.
- d. Entryway design that considers at least three strategies to minimize the ingress or permeation of water into the building.
- e. Installation of a continuous drainage plane (e.g., a weather-resistant barrier integrated with flashing systems at penetrations), interior to the exterior cladding.
- f. Minimization of capillary suction (wicking) in porous building materials through one of the below capillary break methods:
 1. Free-draining spaces (e.g., between exterior cladding, weather-resistant barriers in wall assemblies).
 2. Non-porous materials (e.g., closed-cell foams, waterproofing membranes, metal) between porous materials.

PART 2 DESIGN INTERIORS FOR MOISTURE MANAGEMENT (MAX : 1 PT)

For All Spaces:

1: Condensation and liquid water management

The project implements measures to manage moisture in interior spaces that address, at a minimum, the following:³

- a. Protection of moisture-sensitive building materials and selection of moisture-resistant materials or finishes in surfaces likely to be exposed to liquid water (e.g., finished floors) or that may absorb moisture such as interior sheathing in basements, areas at or below grade, bathrooms, janitorial rooms or kitchens.
- b. Condensation on cold surfaces such as basements, slab-on-grade floors, the inside of exterior walls and glazing.

2: Water leak control in fixtures

The following requirements are met:

- a. All hard-piped fixtures, such as toilets, dishwashers, icemakers, water treatment devices and clothes washers, have a labeled, readily accessible single-throw manual shut-off or automatic shut-off at point-of-connection.
- b. For water treatment devices that have a waste or drain line (e.g., reverse osmosis systems and water softeners), the drain or waste line is plumbed in-place and is equipped with a backflow prevention system such as an air gap or a backflow preventer valve.

PART 3 IMPLEMENT MOLD AND MOISTURE MANAGEMENT PLAN (MAX : 1 PT)

For All Spaces:

1: Operational moisture management

The project implements a moisture management plan for building operations that contains the following:

- a. A schedule of periodic inspections for signs and potential sources of water damage or pooling, discoloration and mold on ceilings, walls, floors and HVAC equipment.³
- b. A system or inspection protocol to periodically assess water pipe leaks.
- c. A system for occupants and tenants to notify building management about mold or water damage.

2: Leaks and mold inspections

The following requirement is met:

- a. Results of inspections for mold and leaks (including any mold test results) are submitted annually through the WELL digital platform.

REFERENCES

1. Institute of Medicine. Damp Indoor Spaces and Health. Washington, D.C.: National Academies Press; 2004.
2. Mudarri D, Fisk WJ. Public health and economic impact of dampness and mold. Indoor Air. 2007;17(3).
3. U. S. Environmental Protection Agency. Moisture Control Guidance for Building Design, Construction and Maintenance. Washington, DC2013.
4. World Health Organization. WHO guidelines for indoor air quality: dampness and mould. 2009.

W08 HYGIENE SUPPORT | O (MAX : 4 PT)

Intent : Ensure availability of bathrooms and support hygienic hand washing and toilet use practices for all individuals.

Summary : This WELL feature requires projects to provide bathrooms that accommodate users with diverse needs and to improve hygiene by offering large sinks, soap containers, hand drying support and reduced touch points.

Issue : All humans share the critical need to access bathrooms. Proper hand hygiene is key to reducing the incidence of gastrointestinal and respiratory diseases after bathroom use.¹ Access to bathrooms can be challenging for some.²⁻⁴ For example, women often lack necessary bathroom accommodations due to an insufficient quantity and lack of sanitary materials.^{2,4} Caregivers, small children, older adults or other individuals who require assistance frequently lack access to facilities that support their needs.² Despite hand washing, hands can only become as clean as the surrounding environment. Sinks may harbor pathogenic bacteria that can migrate onto hands during washing.⁵ Water splashing from the drain may spread bacteria to surrounding areas.⁶⁻⁹ Additionally, soap and the inside of liquid soap containers often remain contaminated after use; thus, best practice and research recommends that soap dispensers not be topped off.^{7,8} Lastly, once an individual's hands are cleaned, they can more easily become re-infected when wet compared to when dry.^{5,10}

Solutions : Bathrooms can be designed and furnished to ease hygiene, particularly for menstruating women (e.g., by supplying female hygiene products)^{2,4} and for supporting caregivers of children¹¹ and individuals with physical and mental disabilities.² Trash receptacles, baby changing stations, large sinks, fragrance-free soap, hand dryers and fixtures that minimize unnecessary contact with the hands can be provided. Single-user facilities with gender neutral signage provide safe, comfortable bathroom availability for individuals of all gender identities.¹² Visual cues that promote hand hygiene may improve compliance with established guidelines in certain segments of the population.¹³⁻¹⁵

PART 1 PROVIDE EQUIPPED BATHROOMS (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Furnishings for all bathrooms

The project meets the following requirements:

- a. All bathrooms include:
 1. Trash receptacles in stalls (in women's and single-user bathrooms). If toilet paper cannot be flushed down toilets, trash receptacles must be placed in all bathroom stalls such that they do not impede wheelchair/mobility aid access.
 2. Sanitary pads, tampons and/or other menstrual products at no cost or subsidized by at least 50% (in women's and single-user bathrooms).
 3. A hook, shelf or equivalent storage support in each toilet stall. For wheelchair accessible stalls, storage support items are placed no higher than 48 in.
- b. All occupants have access to at least one bathroom per occupiable floor that provides a stall that can accommodate a wheelchair user and care attendant.
- c. All occupants have access to at least one bathroom that provides an infant changing table.
- d. All regular occupants may confidentially request a syringe drop box, which is made available at no cost in one or more bathrooms.¹⁶
- e. All single-user bathrooms (if present) are open to all individuals with accompanying signage.
- f. If present, floor drains are equipped with a self-primed liquid-seal trap or a waterless trap seal.¹⁷

2: Furnishings for family bathrooms

For projects where the majority of occupants are visitors (e.g., shopping malls, airports, museums), family bathrooms are provided to meet expected demand by individuals in need of accompaniment or assistance in the bathroom (e.g., children, individuals with mental or physical disabilities) and contain the following amenities:¹⁸

- a. Changing table for infants.
- b. Children's toilet facilities or accommodations for child use of adult-size toilet.
- c. Children's sinks or accommodations for child use of adult-size sink (e.g., availability of stepstool).
- d. Motion sensor lights.
- e. Slip-resistant floors.
- f. Grab bars.
- g. At least one designated location for bags in each stall (e.g., hook, shelf separate from changing table and sink).
- h. Meets the room and stall dimensions required by local code for wheelchair accessibility.

PART 2 PROVIDE HANDS-FREE FIXTURES IN BATHROOMS (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

All bathrooms meet the following requirements:

- a. Toilets are equipped with hands-free flushing.
- b. Contactless soap dispensers and hand-drying accommodations are provided.
- c. Users can exit the bathroom hands-free.
- d. Faucets meet the following:

1. Sensor-activated.
2. Equipped with a programmable line-purge system.
3. If mixing is used, hot- and cold-water lines are mixed at the point of use.

PART 3 SUPPORT EFFECTIVE HANDWASHING (MAX : 1 PT)

For All Spaces:

All sinks where handwashing is expected (e.g., kitchens, bathrooms, break rooms and wellness rooms), meet the following requirements:

- a. The faucet design prevents the water column from flowing directly into the drain or a sink drain stopper is installed.^{6,19}
- b. Water does not splash outside the sink when the faucet is fully open.
- c. Newly installed sinks meet the following design parameters:
 1. The sink basin is at least 23 cm across in the smallest dimension, measured at the point where the user is expected to place hands during hand washing.
 2. The water column from the faucet spout to the basin is at least 20 cm in length (measured along flow of water, even if at an angle).
 3. The water column is at least 7.5 cm away from any edge of the sink.

PART 4 PROVIDE HANDWASHING SUPPLIES AND SIGNAGE (MAX : 1 PT)

Note : Projects may only achieve this part if Part 1, Part 2 or Part 3 is also achieved.

For All Spaces except Dwelling Units & Guest Rooms:

1: Handwashing support

For all sinks where handwashing is expected (e.g., bathrooms, break rooms, food prep and wellness rooms), the following are present within the room:

- a. Fragrance-free liquid hand soap dispensed through one of the following:
 1. Sealed dispensers equipped with disposable soap cartridges.
 2. Dispensers with detachable and closed containers for soap refill. Soap containers must be washed and disinfected when emptied, before refilling.
- b. One of the following methods for hand drying:
 1. Paper towels.
 2. Hand dryers equipped with a HEPA filter. Filter replacement and equipment maintenance are carried out per manufacturer's instructions. This method is not available for healthcare projects.
 3. Fabric hand towel rolls with dispensers, with rolls replaced before reaching their end of service.
 4. Washable fabric hand towels accompanied by a used towel container placed nearby. Towels are washed before reuse.
- c. Signage displaying steps for proper hand washing.

For Commercial Kitchen Spaces & Commercial Dining Spaces:

1: Provide Handwashing Signage in Commercial Kitchens

The following requirement is met:

- a. Clear signage directing toward the nearest handwashing location is present at the entrance to all areas intended for food preparation and consumption.

For Guest Rooms:

The following are provided in each bathroom:

- a. Fragrance-free hand soap via one of the following:
 1. Individually wrapped bar soap, replaced during room turnover, and drainable soap bar racks.
 2. Sealed dispensers of liquid soap equipped with disposable soap cartridges.
 3. Liquid soap dispensers with detachable and closed containers for soap refill. Soap containers must be washed and disinfected when emptied, before refilling.
- b. One of the following methods for hand drying:
 1. Reusable cloth towels, replaced and washed at least during room changeover.
 2. Paper towels.
 3. Hand dryers equipped with a HEPA filter. Filter replacement and equipment maintenance are carried out per

manufacturer's instructions.

4. Fabric hand towel rolls with dispensers, with rolls replaced before reaching their end of service.

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W09 B ONSITE NON-POTABLE WATER REUSE | O (MAX : 2 PT)

Intent : Conserve water through non-potable water systems without compromising the health of the building occupants.

Summary : This WELL feature requires projects to implement a safety plan when capturing and using non-potable water within the project boundary.

Issue : Efficient water management is an ever-growing need for supporting and sustaining human existence.¹ The rise of extreme droughts and subsequent long-term water scarcity are requiring changes in how humans think about and use water.¹ Conversely, severe rain events can overload wastewater treatment plants with storm- and rainwater, potentially resulting in combined sewer overflows (CSOs), in which untreated water is released to natural bodies of water and beaches downstream.² Buildings can reduce the pressure on city-wide water infrastructure by supporting efficient allocation of resources, such as minimizing the use of municipally sourced water for applications that do not require potability (e.g., irrigation, flushing toilets).² However, without proper design and operations management, there is a risk of contamination of potable water lines or oral or respiratory exposure to non-potable water.²⁻⁶

Solutions : The design and operations of a healthy, non-potable water system must address the water sources and uses. For instance, water coming from clothes washers and showers, as well as stormwater collected at the street level, requires treatment and water quality monitoring before reuse. The system should be designed so that water capture and reuse does not create niches of water stagnation, and storage tanks for non-potable water need periodic checks and safeguards against overflows.⁵ Additionally, pipes destined to distribute non-potable water must not be commingled with those for drinking water, and proper control measures must be in place if potable water supplements non-potable water applications (e.g., toilet flushing).² Finally, smells and odors need to be managed to avoid the perception of a health threat, as this may hinder efforts to improve water management and conservation in a building.^{7,8} While there are many factors to consider, these challenges can be managed through a properly implemented safety plan that addresses health endpoints in the collection, treatment and distribution of non-potable water within the building.⁹ The success in the implementation of this plan also depends on third party verification, in which the plan is evaluated with respect to the achievement of its health endpoints.¹⁰ Additionally, education and signage are essential in supporting occupants' understanding of the positive impact of sensible water utilization within buildings and of the measures taken to protect human health.¹¹ By properly managing and controlling non-potable water reuse and capture systems, projects can help maintain occupant health and safety, while contributing to decreased water use by up to 75% in new buildings.¹²

PART 1 IMPLEMENT SAFETY PLAN FOR NON-POTABLE WATER CAPTURE AND REUSE (MAX : 2 PT)

For All Spaces:

1: Safety plan description

The project or organization implements a safety plan that contains the following:⁵⁻⁹

- a. A list of key roles for design, operations, maintenance and third-party inspection of the non-potable water system capture, treatment and use.
- b. A list of all applicable codes and regulations in the jurisdiction where the non-potable water reuse system is being installed and that govern the design, commissioning, and approval of operation of the system.
- c. A process flow diagram that displays the non-potable water sources, conveyances, storage units, treatment devices and points of use, emphasizing the points where makeup potable water (i.e., water needed to supplement non-potable needs) may be added.
- d. A description of the system that includes the sources and estimated contaminant loads of the non-potable water, the intended uses for the non-potable water, the water treatment devices (if any) and their certifications, and the water quality parameters expected at the points of use.
- e. An analysis of how human exposure to pathogens through ingestion and inhalation of non-potable water is minimized, including (if applicable) a description of how the potable water network is protected from the introduction of non-potable water, emphasizing strategies that address cross-connection control and backflow prevention.
- f. A description of the signage and identifiable pipe color-coding to distinguish the non-potable water network.
- g. A list of strategies for the control of odors, nuisances and vectors due to stagnation of non-potable water.
- h. The provisions for emergency operations caused by overflow of storage tanks, leaks and outages.
- i. A list of operational parameters (such as flow, turbidity, coliforms or other treatment-dependent indicators) to monitor the intended functioning of the water system, their monitoring frequency and control actions if such parameters are beyond target ranges.
- j. A list of control points where the operational parameters are being measured.
- k. A list of routine maintenance protocols and schedules.
- l. A description of the procedures for system startup, determination of protocols for verification of the safety plan, including Legionella testing if a risk of inhalation exists, and schedule for third-party inspections.

2: Safety plan implementation

Documentation of the following are submitted annually through the WELL digital platform:

- a. Startup procedure.
- b. Maintenance logs.
- c. Results from verification tests (when applicable).
- d. Third-party inspections.

3: Safety plan communications

Conspicuous signage is present to communicate the following:

- a. To distinguish potable from non-potable water (where applicable).
- b. To highlight the safety features and conservation goals of the non-potable water system.

Note :

Projects are not required to provide information on the individuals assigned to perform the key roles determined in the safety plan.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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